7.11 ELECTRICAL SYSTEM

The electrical system provides 28V DC electrical power to the helicopter Starter, Main and Auxiliary Buses. An onboard 24V DC nickelcadmium battery supplies engine starting power and serves as an inflight emergency power source should both generators fail.

7.11.1 **Power Supply** (see Fig. 7-17)

The power supply comprises two engine-driven 150-ampere DC starter/generators. The two generators, each with a voltage regulator and a differential reverse current relay, operate in parallel to supply 28V DC single-pole negative ground electrical power. Each voltage regulator controls the output voltage of its generator and equalizes the load between the two generators. The differential reverse current relay monitors generator output and will disconnect the generator from the system if its output voltage is less than the battery voltage and will cause the respective **GEN** warning light to come on.

7.11.2 **Distribution** (see Fig. 7-17)

Electrical power is distributed through use of three primary buses:

– Starter Bus primary power distributer that supplies battery and/or external power for engine starting, generator power for equipment requiring high current loads and all sources of power for further distribution to other buses.

– Main Bus supplies power to all equipment and systems essential for normal helicopter operation. This bus further supplies two Radio Nav Buses that power the COMM/NAV equipment.

– Auxiliary Bus supplies power to all non-essential consumers only when both generators are operational or when an external power source is connected.

7.11.3 **Controls, Indicators and Annunciators** (see Fig. 7-17)

STARTER/GENERATOR, ENG 1/ENG 2 switches when placed in the GEN ON position allow control voltages from the respective generator to flow to the reverse current relays. When the control voltage is greater than the battery voltage the relay will switch the generator on-line and the appropriate **GEN** warning light will go off. During external power assisted starts, the generators are not brought on-line regardless of STARTER/GENERATOR switch position as long as the external power source is providing power to the system.

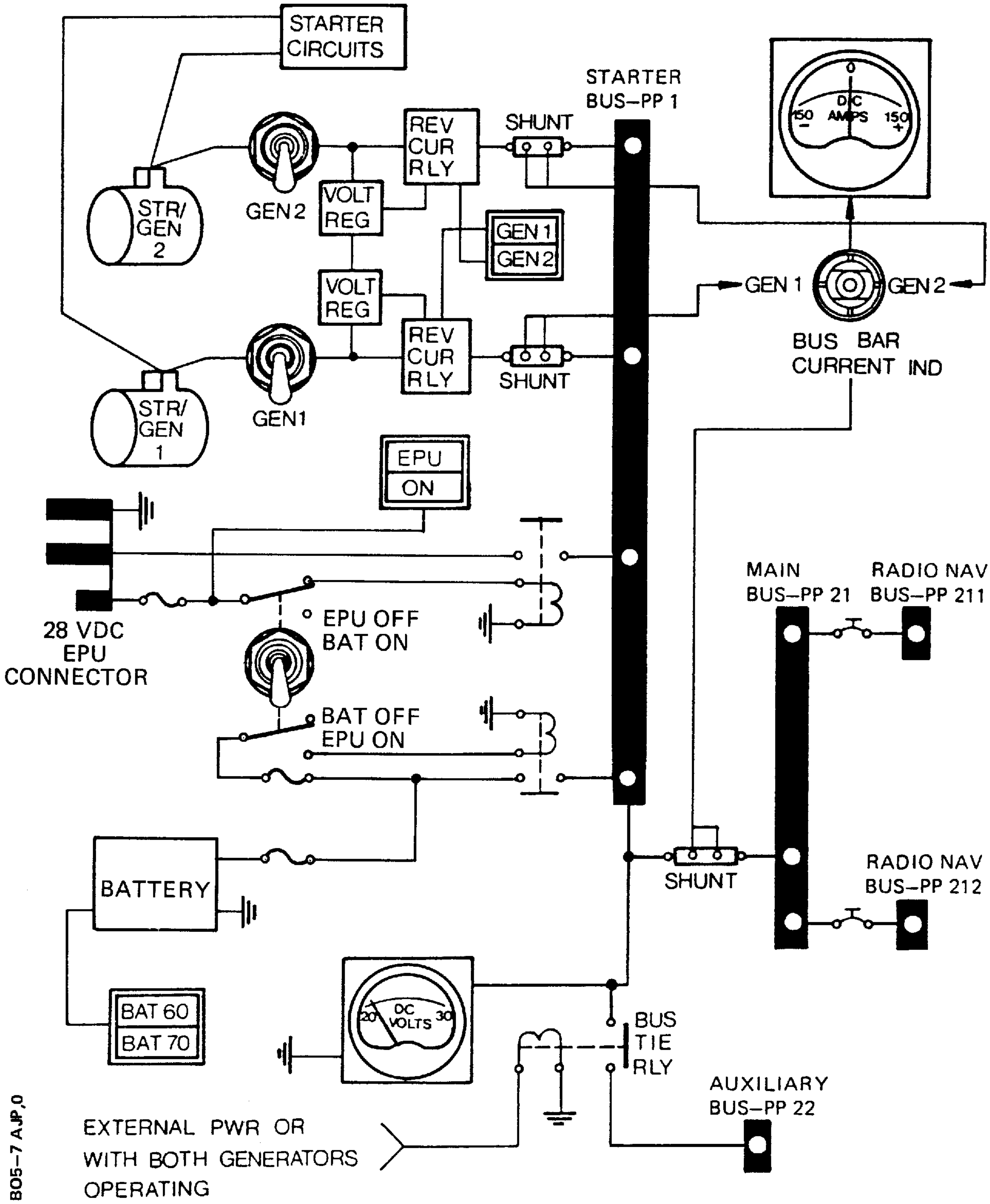
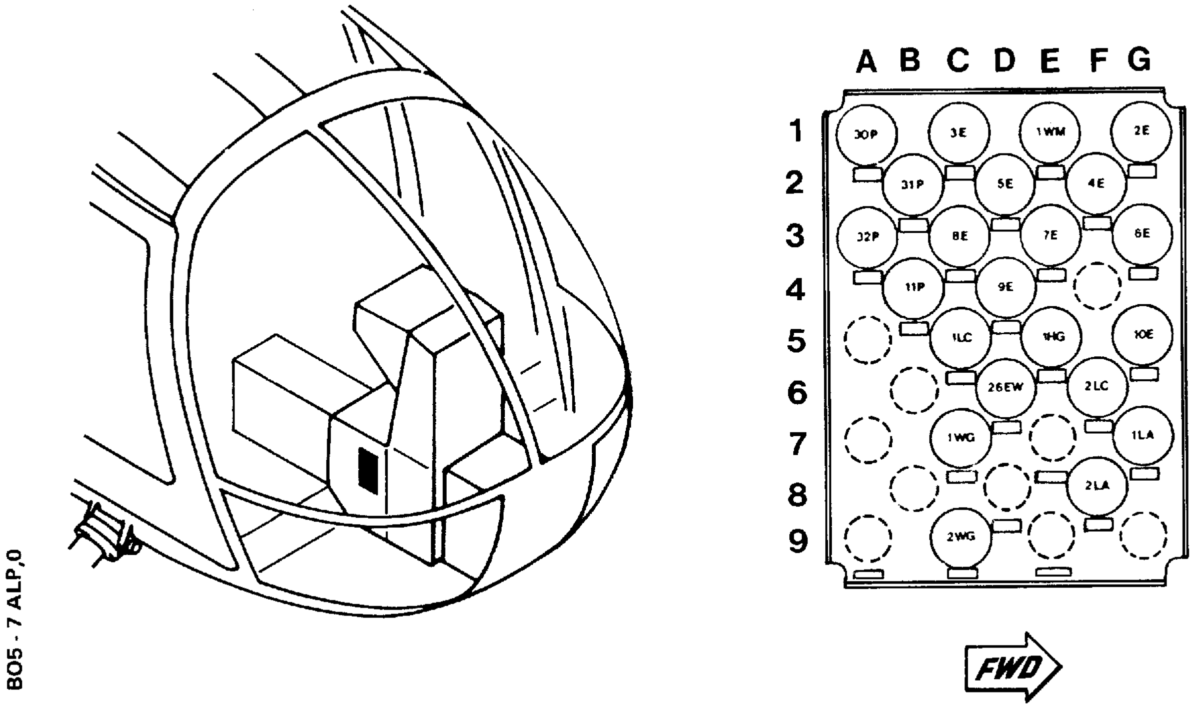


Fig. 7-17 Electrical System Schematic



FUSE IDENT	PANEL LOCATION	RATING	CIRCUIT(S) PROTECTED
2E	1G	1A	LOW FUEL warning
3E	1C	1A	Fuel quantity indicator
4E	2F	1A	RPM warning system
5E	2D	1A	Triple oil pressure indicator
6E	3G	1A	Triple oil temperature indicator
7E	3E	1A	Fuel pressure indicator
8E	3C	4A	Warn lt test, T PLUG and OIL COOL warnings (optional)
9E	4D	1A	Fuel FILT 1/FILT 2 warnings
10E	5G	1A	Engine MAG PLUG 1/MAG PLUG 2 warnings
26EW	6D	1A	Transmission T OIL warning
1HG	5E	3A	Pitot tube heating
1LA	7G	10A	Anticollision lights
2LA	8F	4A	Position lights
1LC	5C	3A	Instrument lights
2LC	6F	4A	Interior lights and utility light
11P	4B	4A	Battery control relay
30P	1A	4A	EPU control and EPU ON caution light
31P	2B	4A	Ammeter main bus shunt - negative pole
32P	3A	4A	Ammeter main bus shunt - positive pole
1WG	7C	1A	Fire warning - Engine 1
2WG	9C	1A	Fire warning - Engine 2
1WM	1E	2A	Mast moment indicator system

Fig. 7-18 Typical Fuse Panel Layout